

MIS 311 – Introduction to Business Analytics

Student Name	Nguyen Ngoc Mai Trang
Subject	MIS 311 – Introduction to Business Analytics
Assessment	Assignment #1: Personal Portfolio Site
Dataset	Supermarket Sales
Tools Used	Python – pandas, matplotlib, seaborn

1. Data Overview

1.1 Dataset Background

The **Supermarket Sales** dataset captures transaction-level records from a retail supermarket chain operating across two branches (A and B) in the United States. It is used to study consumer behaviour, branch performance, and product category trends in a retail business context.

Rows (raw)	253
Columns	8
Source	Retail supermarket transaction records
Branches	Branch A (New York / Chicago), Branch B (Los Angeles)

1.2 Column Descriptions

Column	Type	Description
sale_id	object	Unique transaction identifier
branch	object	Supermarket branch (A or B)
city	object	City where the branch is located
customer_type	object	Type of customer: Member or Normal
product_name	object	Name of the product sold
product_category	object	Category the product belongs to
quantity	int	Number of units sold per transaction
total_price	float	Total transaction value in USD

2. Data Cleaning

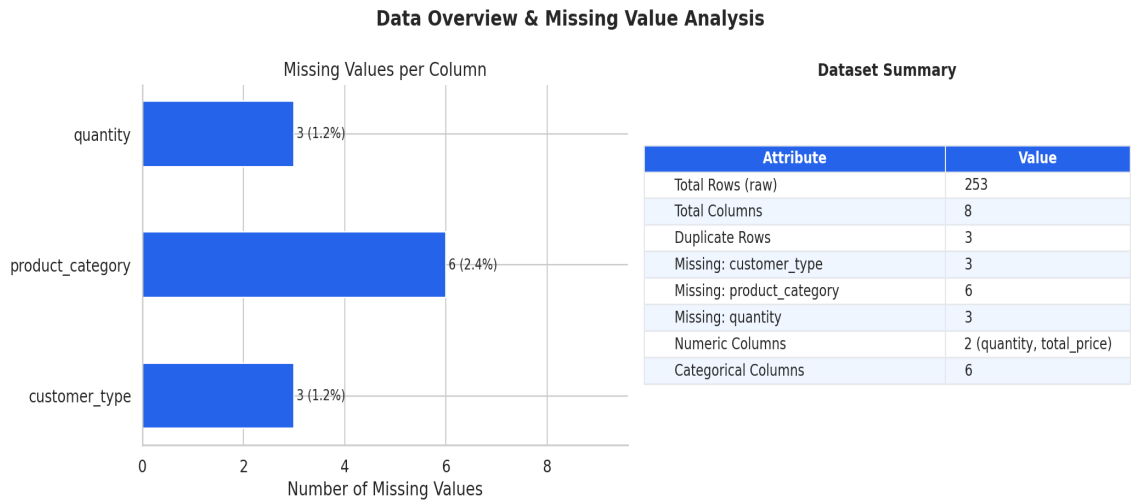


Figure 1: Missing value analysis and dataset summary

2.1 Missing Values

Three columns contained missing values:

- **customer_type**: 3 missing values (1.2%) — filled with 'Unknown'.
- **product_category**: 6 missing values (2.4%) — filled with 'Unknown'.
- **quantity**: 3 missing values (1.2%) — filled with the **median** value (11.0) to avoid skewing the distribution.

2.2 Duplicate Rows

3 duplicate rows were identified and removed from the dataset.

Issue	Rows Affected Action
Missing customer_type	3 rows — Filled with "Unknown"
Missing product_category	6 rows — Filled with "Unknown"
Missing quantity	3 rows — Filled with median (11.0)
Duplicate rows	3 rows — Removed
Final clean dataset	250 rows x 8 columns

3. Descriptive Statistics

Descriptive Statistics - Numerical Variables

Statistic	Quantity (units)	Total Price (USD)
Count	250.00	250.00
Mean	10.62	124.19
Std Dev	5.99	102.98
Min	1.00	2.18
25%	5.00	40.89
50% (Median)	11.00	95.43
75%	16.00	194.94
Max	20.00	427.14

Figure 2: Descriptive statistics for numerical variables

Distribution of Key Numerical Variables

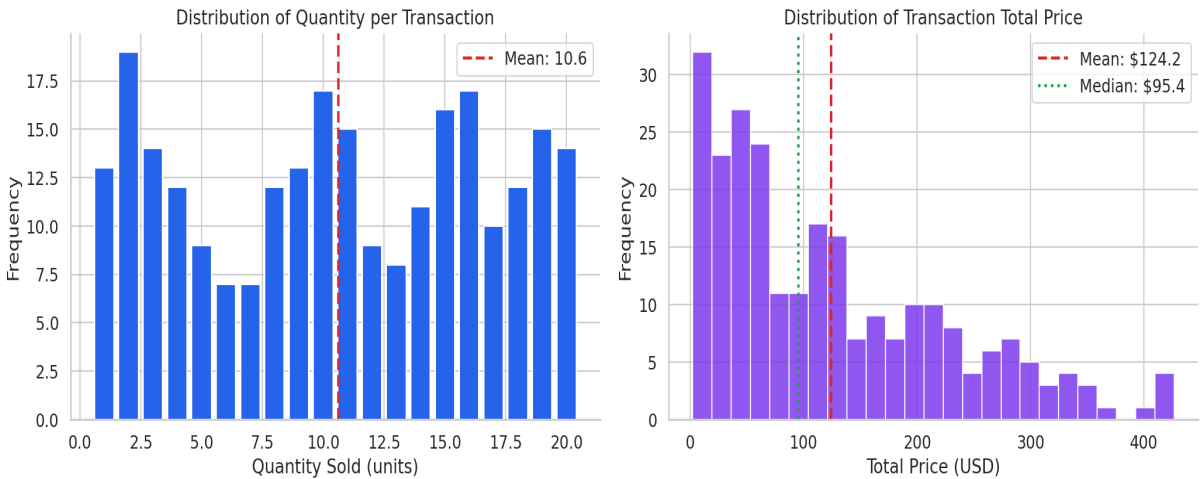


Figure 3: Distribution of Quantity and Total Price

4. Key Insights

Insight 1: Fruits and Beverages Are the Top Revenue-Generating Categories

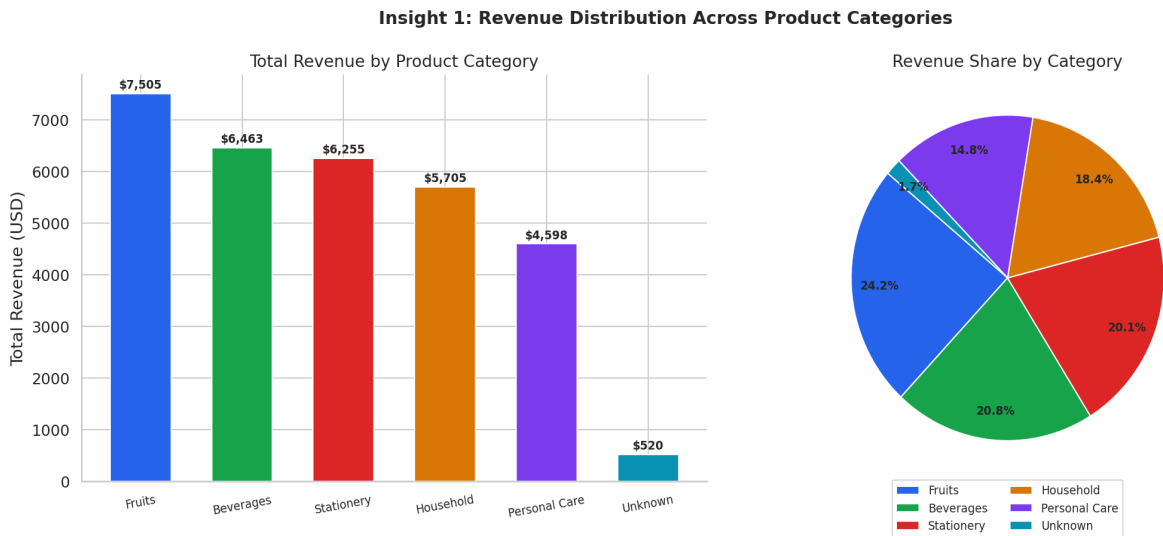


Figure 4: Total revenue by product category

Fruits generated the highest total revenue at **\$7,505.08**, accounting for approximately **23.8%** of all sales, followed by Beverages at **\$6,463.17** (20.5%). Together, these two categories contribute over **44%** of total revenue, making them the most commercially important product lines in the store.

Business Implication: Management should prioritise consistent stock availability and promotional activities for Fruits and Beverages. These are likely high-frequency purchase items that drive repeat customer visits. Investing in fresh produce quality and beverage variety could further strengthen revenue from these top-performing categories.

Insight 2: Branch A Significantly Outperforms Branch B, and Members Spend More

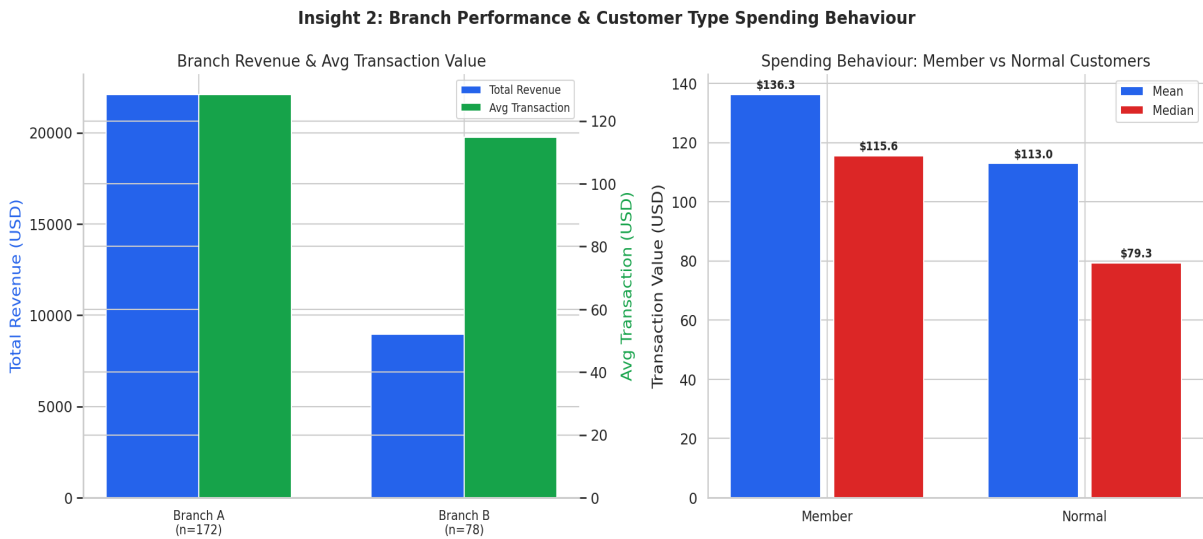


Figure 5: Branch performance and Member vs Normal customer spending

Branch A recorded total revenue of **\$22,097.47**, which is nearly **2.5 times higher** than Branch B at **\$8,948.81**. Additionally, Member customers have a higher mean transaction value of **\$136.30** compared to Normal customers at **\$112.99** — a difference of approximately **20.6%**.

Business Implication: Branch A's strong performance suggests it benefits from a higher customer volume or a more favourable location (New York / Chicago). Branch B management should investigate strategies to increase foot traffic and transaction values. The membership gap also highlights the value of loyalty programmes — converting Normal customers to Members could meaningfully increase average spending per visit.

5. References

- Kaggle. (2019). *Supermarket sales dataset*.
<https://www.kaggle.com/datasets/aungpyaeap/supermarket-sales>
- McKinney, W. (2022). *Python for data analysis* (3rd ed.). O'Reilly Media.
- Pandas Development Team. (2024). *pandas documentation* (v2.2). <https://pandas.pydata.org/docs/>
- Waskom, M. (2021). Seaborn: Statistical data visualization. *Journal of Open Source Software*, 6(60), 3021.
<https://doi.org/10.21105/joss.03021>